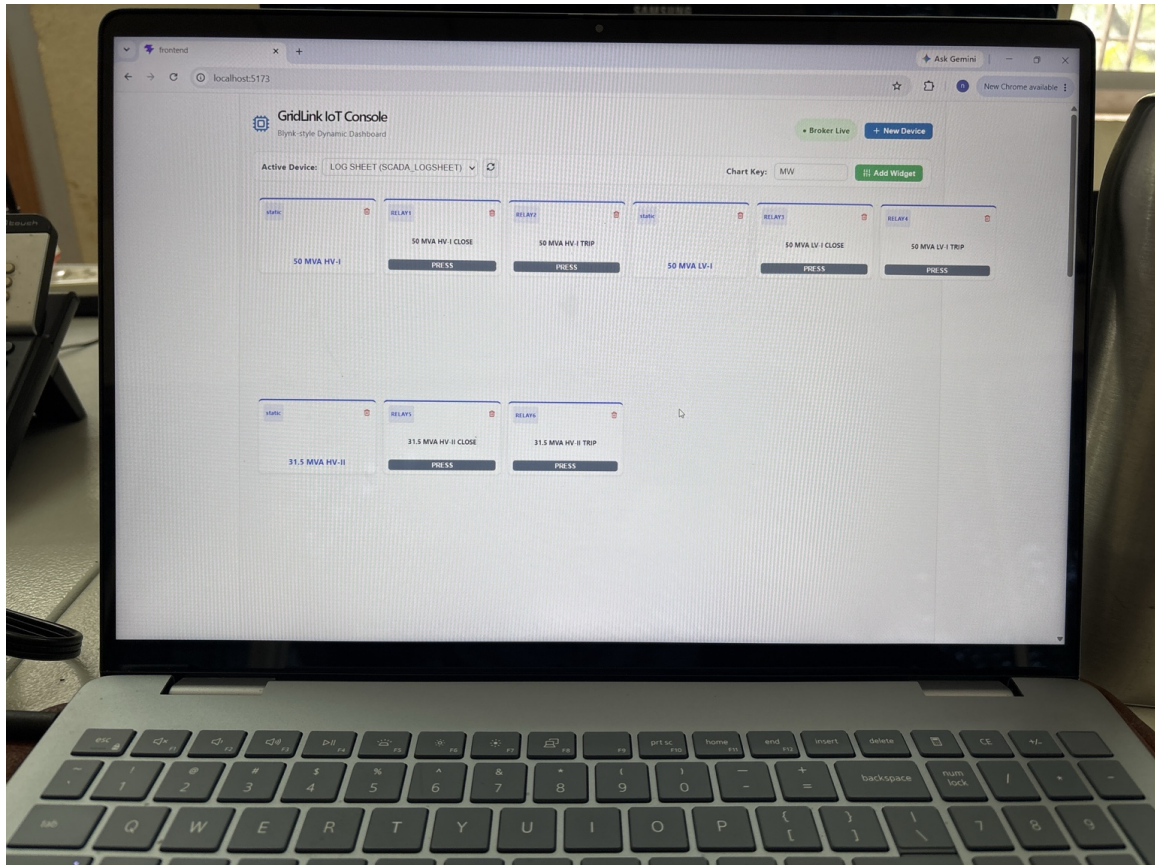


GridLink IoT Console

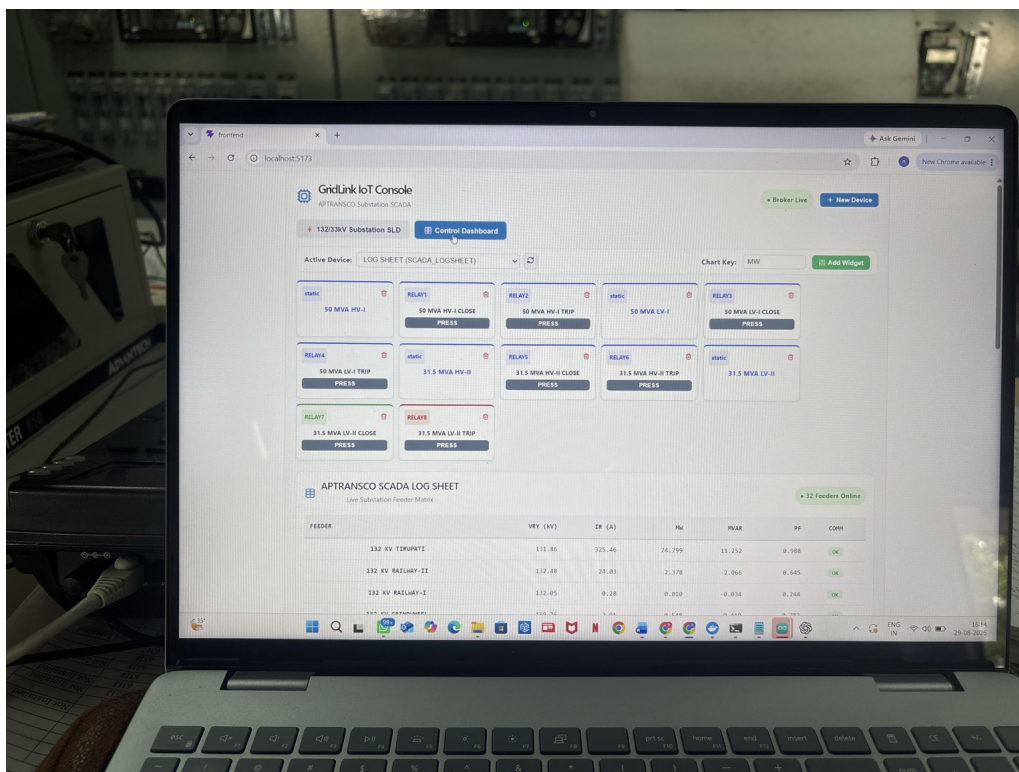
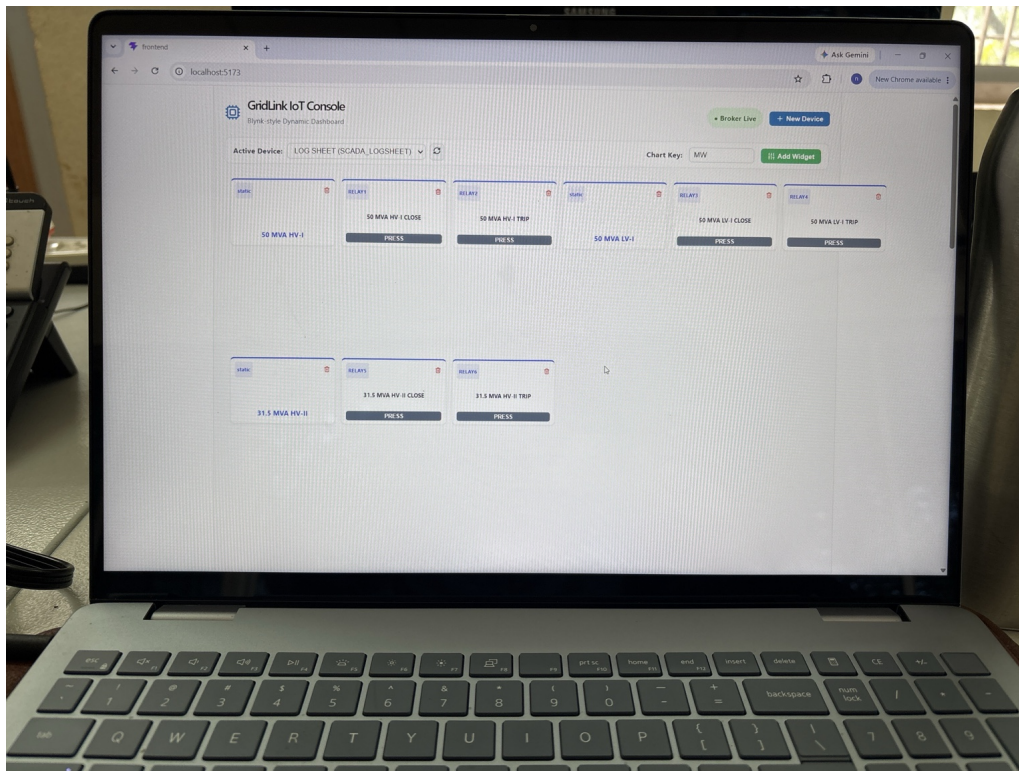
Self-Hosted IoT & SCADA Platform



Project Overview

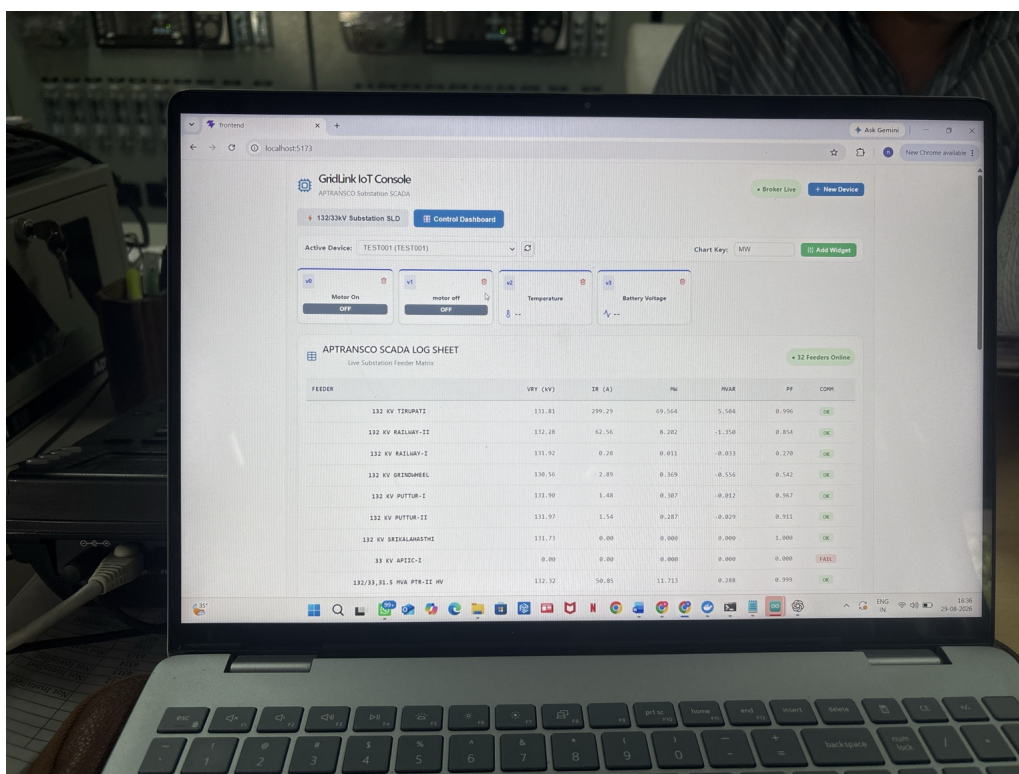
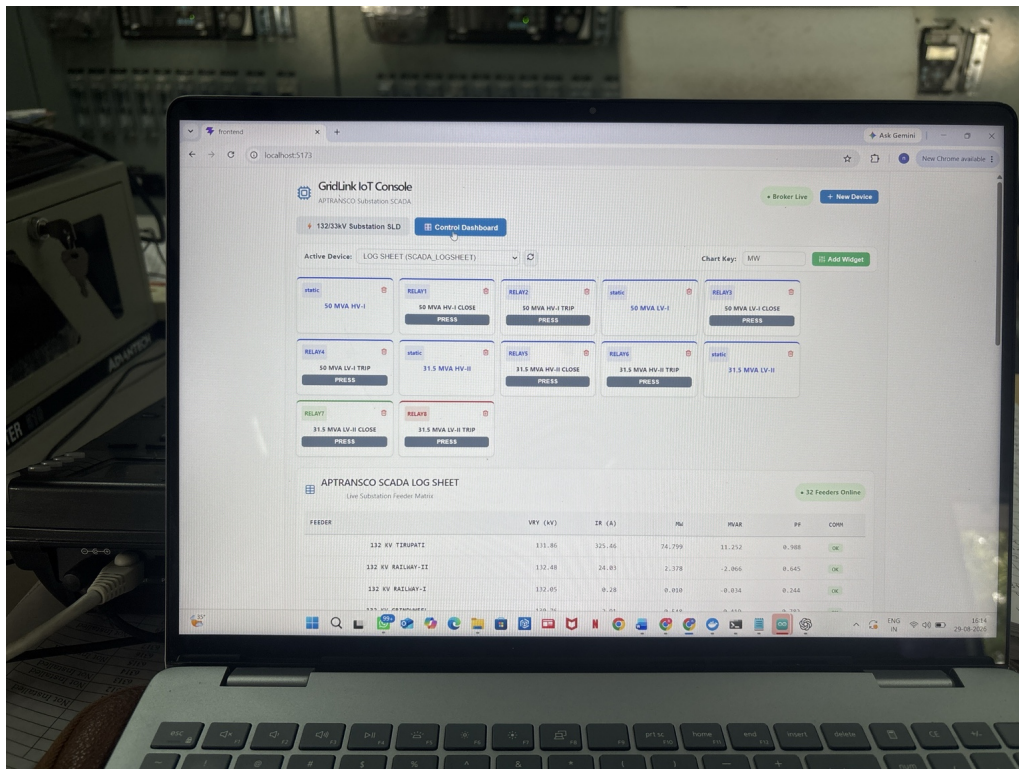
GridLink IoT Console is an in-house Blynk-style IoT platform for industrial monitoring, telemetry, control and SCADA-oriented substation visualization.

GridLink Control Console



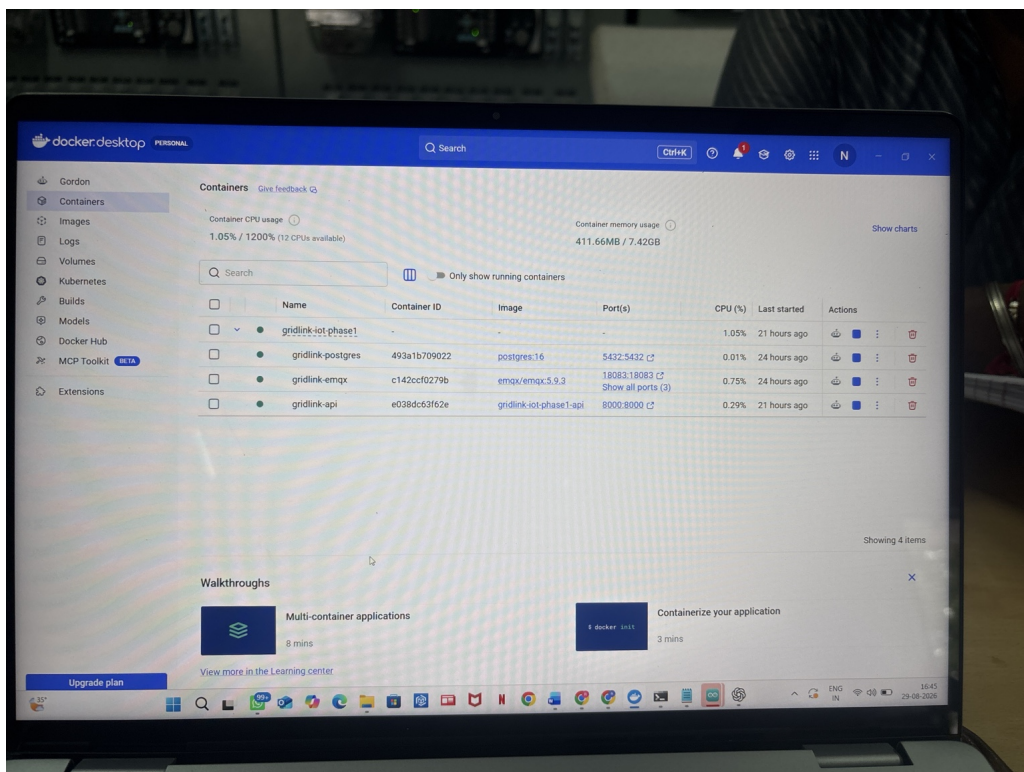
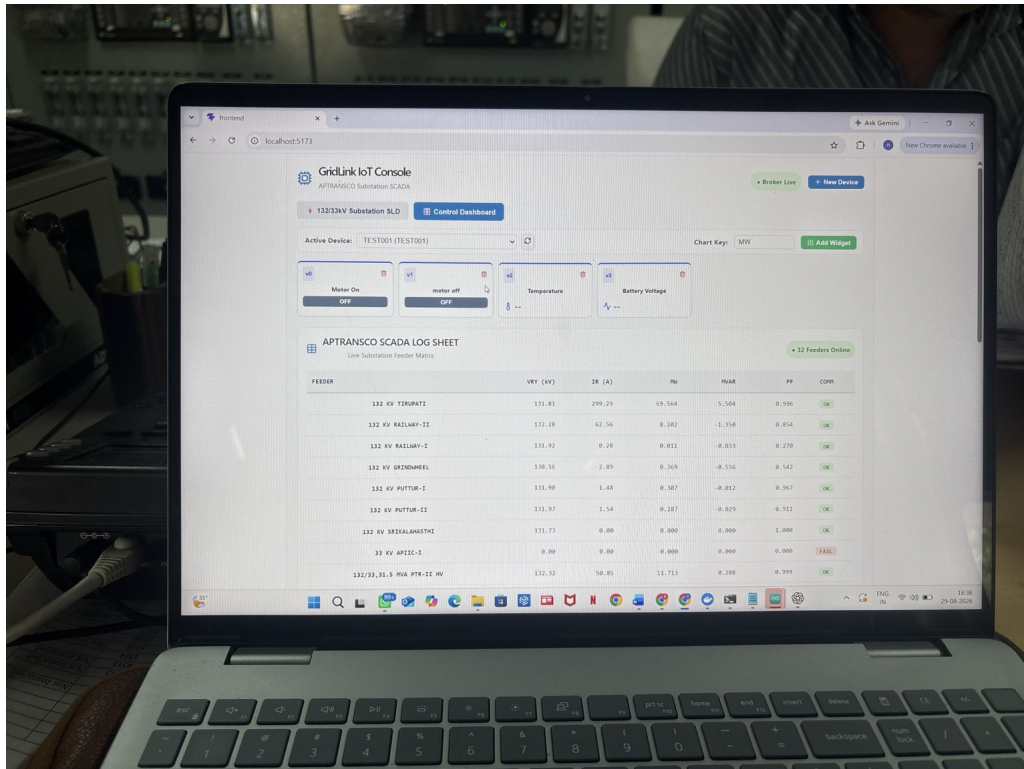
Actual GridLink project interface / integration view.

132/33 kV SCADA / SLD



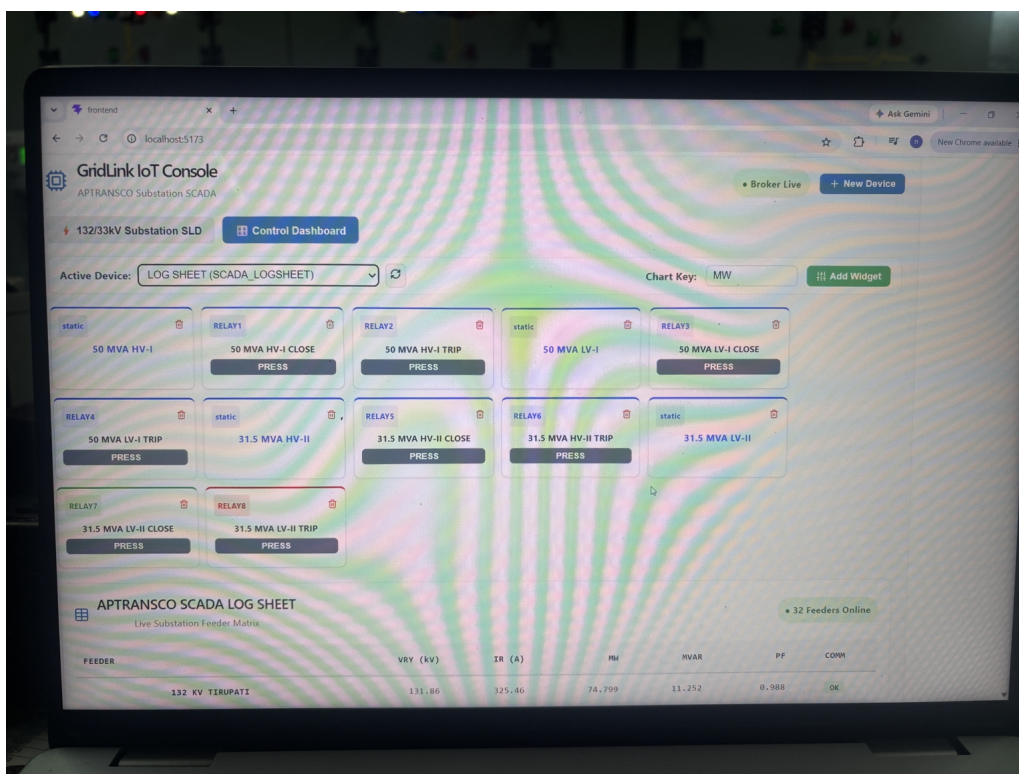
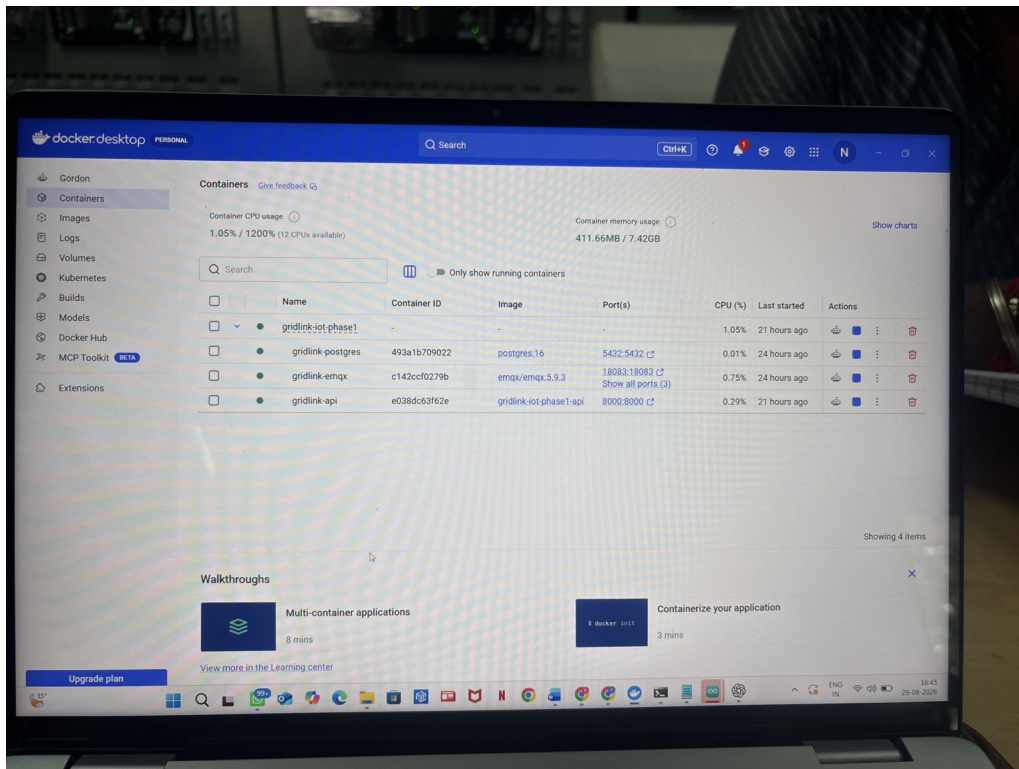
Actual GridLink project interface / integration view.

Live Feeder Monitoring



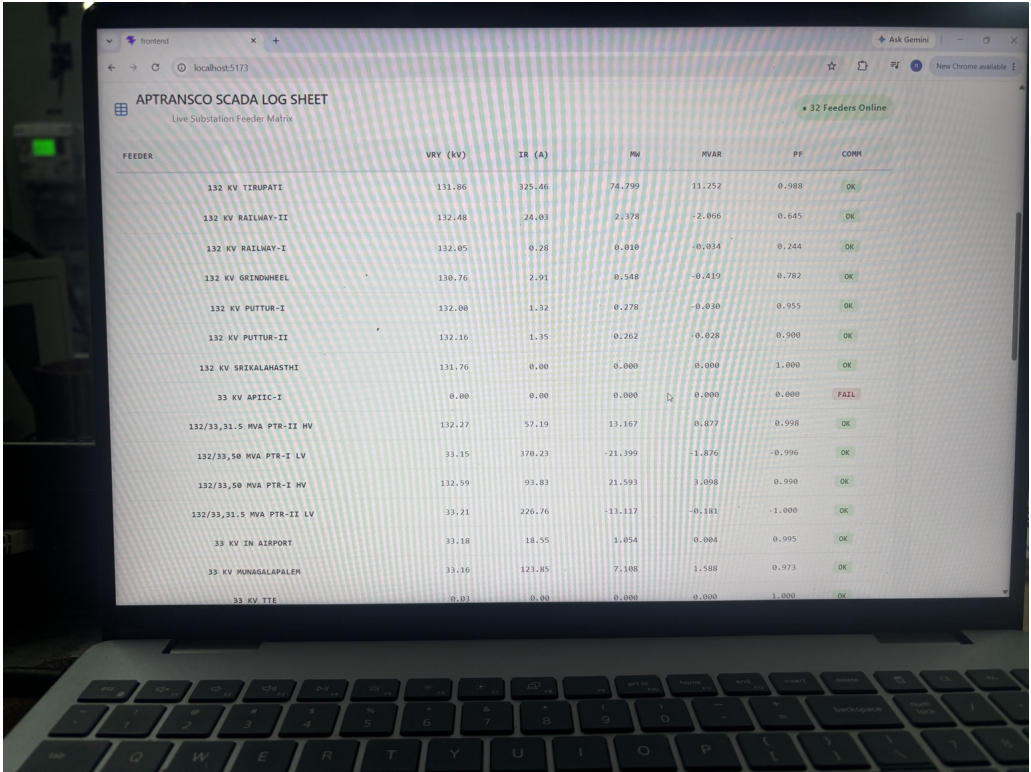
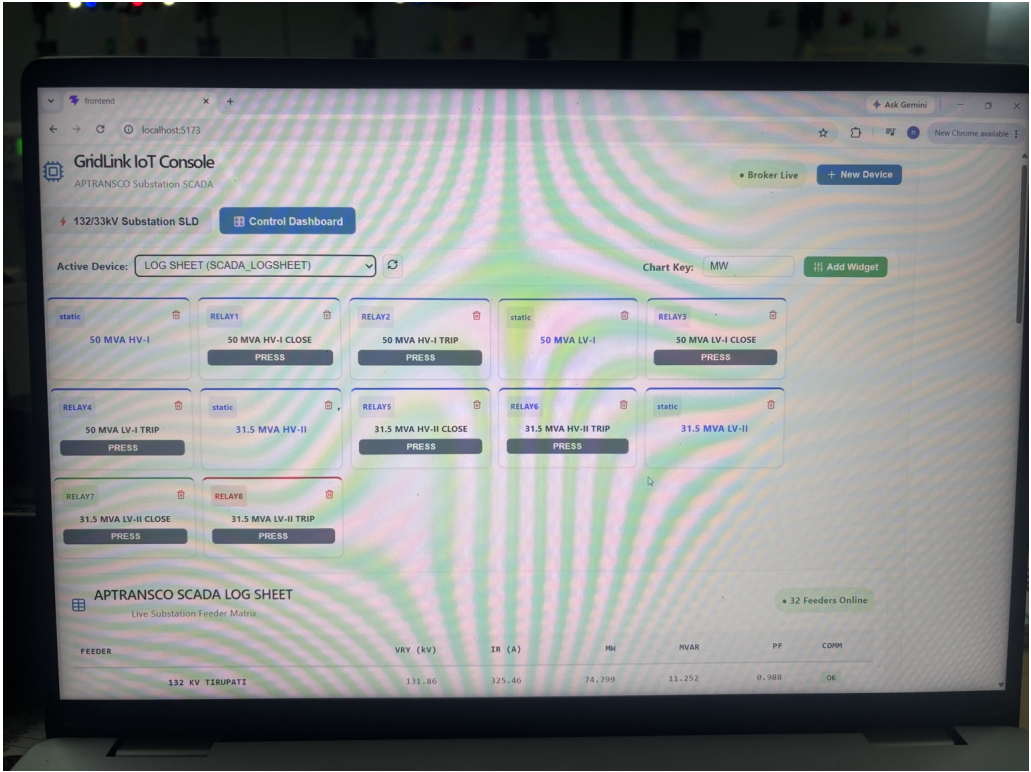
Actual GridLink project interface / integration view.

GridLink Widgets



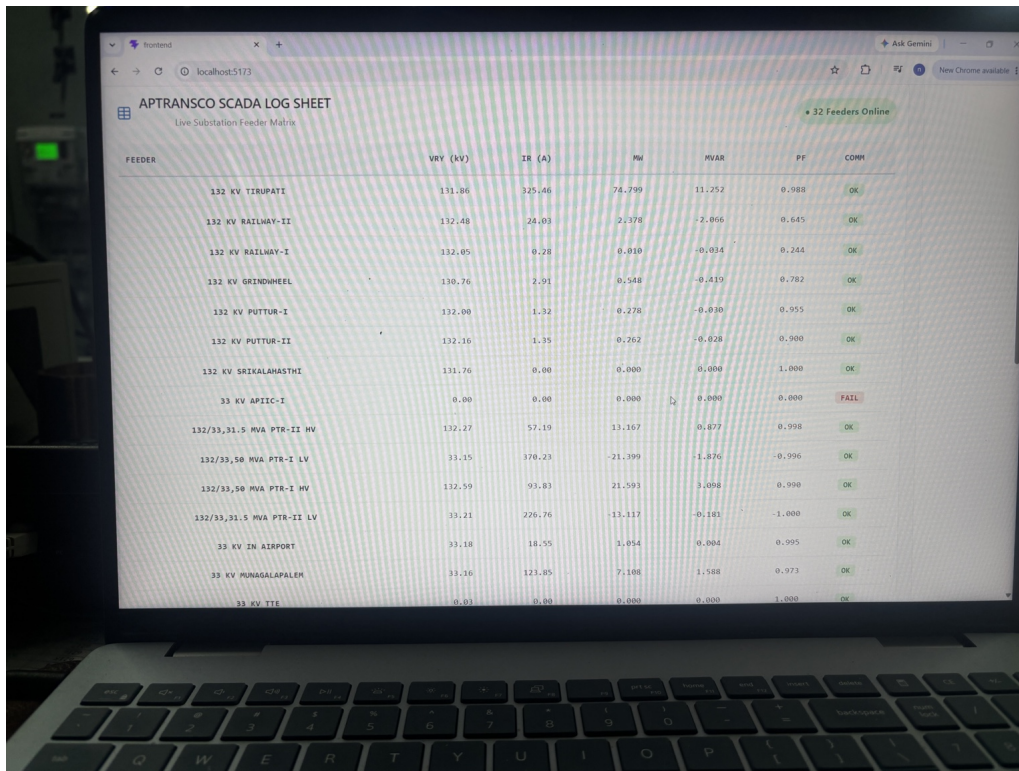
Actual GridLink project interface / integration view.

Field Device Integration

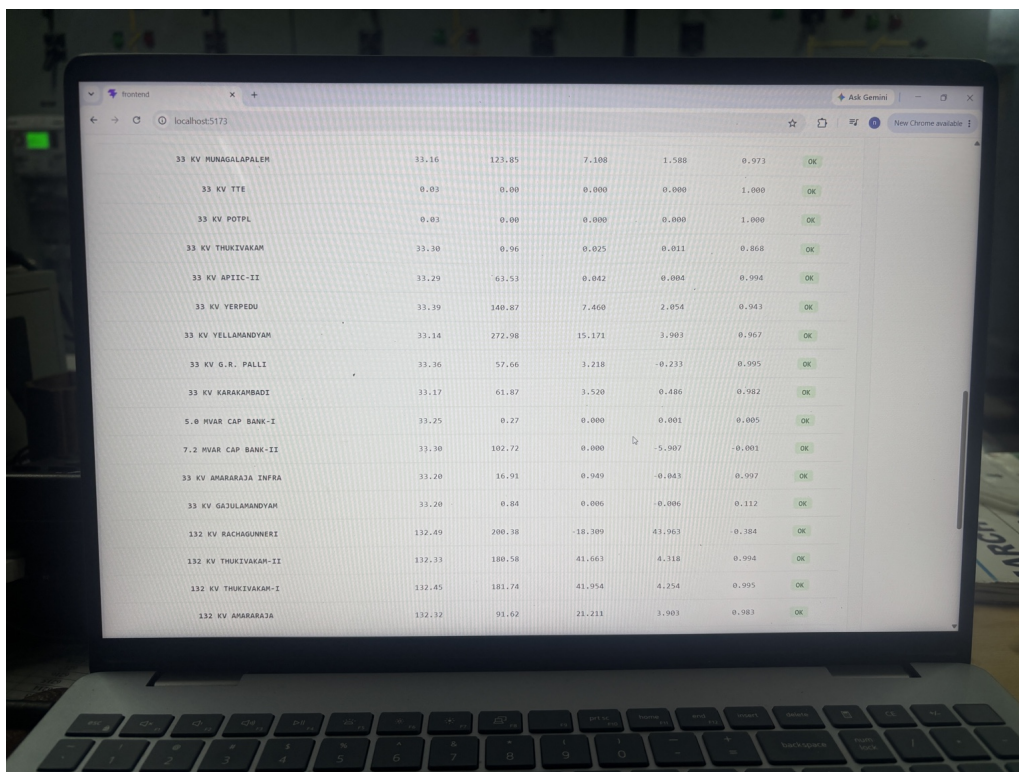


Actual GridLink project interface / integration view.

SCADA Telemetry



FEEDER	VRY (KV)	IR (A)	MW	MVAR	PF	CORR
132 KV TIRUPATI	131.86	325.46	74.799	11.252	0.988	OK
132 KV RAILWAY-II	132.48	24.03	2.378	-2.866	0.645	OK
132 KV RAILWAY-I	132.05	0.28	0.010	-0.034	0.244	OK
132 KV GRINDWHEEL	130.76	2.91	0.548	-0.419	0.782	OK
132 KV PUTTUR-I	132.00	1.32	0.278	-0.030	0.955	OK
132 KV PUTTUR-II	132.16	1.35	0.262	-0.028	0.900	OK
132 KV SRICALAHASTHI	131.76	0.00	0.000	0.000	1.000	OK
33 KV APIIC-I	0.00	0.00	0.000	0.000	0.000	FAIL
132/33,31.5 MVA PTR-II HV	132.27	57.19	13.167	0.077	0.998	OK
132/33,50 MVA PTR-I LV	33.15	170.23	-21.399	-1.476	-0.996	OK
132/33,50 MVA PTR-I HV	132.59	93.83	21.593	3.098	0.990	OK
132/33,31.5 MVA PTR-II LV	33.21	226.76	-13.117	-0.181	-1.000	OK
33 KV IN AIRPORT	33.18	18.55	1.054	0.004	0.995	OK
33 KV MUNAGALAPALEM	33.16	123.85	7.108	1.588	0.973	OK
33 KV TTE	0.03	0.00	0.000	0.000	1.000	OK



33 KV TTE	0.03	0.00	0.000	0.000	1.000	OK
33 KV POTPL	0.03	0.00	0.000	0.000	1.000	OK
33 KV THUKIVAKAM	33.30	0.96	0.025	0.011	0.866	OK
33 KV APIIC-II	33.29	63.53	0.042	0.004	0.994	OK
33 KV YERPEDU	33.39	340.87	7.460	2.054	0.943	OK
33 KV YELLANDYAM	33.14	272.08	15.171	3.903	0.967	OK
33 KV G.R. PALLI	33.36	57.66	3.218	-0.233	0.995	OK
33 KV KARAKAMBADI	33.17	61.87	3.520	0.486	0.962	OK
5.0 MVAR CAP BANK-I	33.25	0.27	0.000	0.001	0.005	OK
7.2 MVAR CAP BANK-II	33.30	102.72	0.000	-5.907	-0.001	OK
33 KV AMARARAJA INFRA	33.20	16.91	0.949	-0.043	0.997	OK
33 KV GATULAMANDYAM	33.20	0.84	0.006	-0.006	0.112	OK
132 KV RACHAGUNNERI	132.49	200.38	-18.309	43.963	-0.384	OK
132 KV THUKIVAKAM-II	132.33	180.58	41.663	4.318	0.994	OK
132 KV THUKIVAKAM-I	132.45	181.74	41.954	4.254	0.995	OK
132 KV AMARARAJA	132.32	93.62	25.213	3.903	0.983	OK

Actual GridLink project interface / integration view.

System Architecture

ESP32 / Field Devices → RS485 / Modbus → MQTT / EMQX → FastAPI + PostgreSQL → React GridLink Console

Technology Stack

React + Vite • FastAPI / Python • PostgreSQL • MQTT / EMQX • ESP32 • RS485 / Modbus RTU • Docker / Docker Compose

GitHub Repository

GridLink-IoT-Console with frontend, backend, firmware, documentation, Docker configuration and environment templates.